



SDEV2501

Day 09 – Intro to Data Wrangling

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

Today's objective

- Define **data cleaning** and **wrangling**, and explain their role in data science
- Identify and correct **common issues** in datasets (missing values, duplicates, inconsistent formats)
- Use **data structures** to store and manipulate data
- Apply loops or vectorized operations to **clean data efficiently**
- Leverage **open-source tools** to clean and reshape data
- **Combine data** from multiple data sets

Agenda

- Review **notebooks**
- Discuss steps of data wrangling
- Practice

REVIEW

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TO ALBERTA**



Review

- How to **open** and **run** a notebook

DATA CLEANING

**HAVE YOU EVER MADE A DECISION
BASED ON INCORRECT DATA?**

What is data validation?

- It's a continuation of our conversation about data quality:
 - Data quality asks “is this data good?”
 - Data validation asks “**how** do we keep it good?”
- **Data validation** is the **rules** that prevent invalid data.
 - During data entry
 - During import
 - Before analysis
- Validation reduces **rework, errors, risk, and mistrust.**
- Validation != cleaning.
 - Validation **prevents**, cleaning **repairs.**

Common Validation Techniques



Range Checks

Ensure values are **reasonable**.
e.g. age, percentages, response times



Completeness Checks

Ensure **required** data exist.



Consistency Checks

Ensure the data **agrees** with itself.
e.g. dates make sense, categories align,
totals match components

Validation Tradeoffs

Too strict: valid edge cases are blocked.

Too loose: invalid cases are allowed.

DATA QUALITY

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What do we mean by data quality?

- Brainstorm terms as a class

good analysis +

bad data =

bad decisions

Data Quality != Clean Data

Clean doesn't mean **useful**. **Formatted** doesn't mean **accurate**.
Can we spot the hidden problem in these examples?

1. A nicely formatted spreadsheet shows **average** customer satisfaction = 4.2/5.
2. A customer table has no **duplicates**. All IDs are **unique**.
3. Dates are consistently **formatted** as YYYY-MM-DD.
4. Every cell has a **value**.
5. Categories are **standardized**.

Data Quality

Data quality is defined by how well the data serves its purpose.

- Context is key: one dataset might be useful for one question and awful for another.

Common issues include:

- Missing values
- Duplicates
- Outdated information
- Inconsistent formats

ONE DATASET, 2 QUESTIONS

Let's say we have a dataset of Edmonton businesses containing:

- business type
- Neighbourhood
- number of employees
- opening date
- postal code
- business status

Data issues:

- 15% missing postal codes
- some businesses have outdated status
- opening dates are inconsistent
- neighbourhood names have inconsistent spelling

ONE DATASET, 2 QUESTIONS

- 1) "Where are businesses concentrated geographically?"
 - 2) "How many businesses are currently operating?"
- Which issues matter more for each question?

Missing values

- Blank
- Unknown
- Null

Duplicate values

- Impacts summary values (e.g. totals, trends)
- e.g. double-counted customers

Outdated information

- Was correct at some point – but is no longer relevant
- Easy to find in data portals

Inconsistent Formats

- Dates, currencies, categories...

Practice

customer_id	name	age	signup_date	province	satisfaction
1001	Alex Chen	24	2026-09-01	Alberta	4
1002	Sam Lee	217	09/02/26	AB	5
1003	Jamie Smith		2026-09-03	Alberta	3
1003	Jamie Smith	31	2026-09-03	Alberta	3
1004	Taylor Jones	-4	2026-09-04	Alberta	6
1005	Morgan Brown	42	2026-09-05	alberta	4

DIMENSIONS

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Dimensions of Data Quality



Accuracy



Completeness



Consistency



Timeliness



Validity

Where do bad data come from?

Manual entry

System changes

Merging datasets

Poor definitions

DATA QUALITY REVIEW

- Look for:
 - Missing values
 - Duplicates
 - Typos / inconsistencies
 - Suspicious values
- Debrief:
 - Which issues stood out?
 - Which were hardest to spot?

Fixing vs Flagging



We are focussed on **diagnosis**, not **treatment**.



IRL, sometimes we **fix**, sometimes we **document**, sometimes we **exclude** the dataset. **Judgment** is key.

FIX, FLAG, OR FORGET?



FIX

We have enough information to correct it.



FLAG

We can't safely correct it, but we should identify it.



EXCLUDE

This record/dataset can't reasonably be used for this analysis.



LEAVE

It looks unusual, but there's a legitimate explanation.

- A. A customer's age is recorded as 217.
- B. A survey respondent leaves their income blank.
- C. A transaction has a negative dollar amount.
- D. Customer cities show as Edmonton, Edmonton AB, and EDMONTON.
- E. A dataset contains 40% "Other" in a category column.

Other common steps

- **Rename** columns
 - e.g. "cust_id_1" → "customer_id", "Total Sales " → "total_sales"
- **Imputation**
 - e.g. Fill missing age with the median, or missing category with "Unknown"
- **Normalization** (not the database term!)
 - e.g. Min-max height to 0–1, or z-score income
- **Transformation**
 - e.g. Log-transform skewed prices, one-hot encode "city", extract month from a date

NEXT DAY

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Guided Practice

- Load a data set with pandas
- Inspect the data set (head, info, describe)
- How to
 - handle missing values
 - remove duplicates
 - rename columns
 - convert data types
 - create new columns
 - transform
 - merge datasets
 - normalize

Pair Practice

- As a class, discuss how to clean a given messy data set
- In pairs
 - Clean dataset
 - Document what changed (and why)
- As a class, debrief

Pair debugging

- One person hides a data issue and the other person has to find it

Closing

- 1-minute reflection:
 - Why is data wrangling essential before analysis?

What's next?

- Data visualization

